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Targeted sanctions, resource substitution, and violence against civilians: Localized evidence from African states☆

Jerry Urtuzuastigui, Ore Koren [1,*]

Department of Political Science, Indiana University Bloomington, United States

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ABSTRACT

Since the 1990 s, the UN Security Council increased its use of targeted sanctions, yet we know very little about their unintended impact on civilian victimization, especially at the local level. This study argues that imposing sanctions on armed actors may compel them to seize non-sanctioned agricultural resources to replenish lost revenues and use violence to facilitate appropriation. A new dataset on sanctioned group violence in agricultural areas is developed by matching UN sanctions data with information on local attacks on civilians from the UCDP GED and combining these data with a new geographic dataset. Quantitative analyses establish that monthly changes in cash crop productivity affect political violence by sanctioned actors in Africa, while violence by unsanctioned and pre-sanctioned actors is unaffected. A short case study of Somalia validates the hypothesized mechanism by showing that al-Shabaab actively engaged in violence over such substitution dynamics.

1. Introduction

In 2012, the UN Security Council imposed a sanction targeting charcoal exports on al-Shabaab, a Somalian rebel group, to diminish its most important revenue stream and thus undermine its capacity to militarily challenge the government of Somalia. As its income from the charcoal trade declined, however, al-Shabaab quickly adapted to the sanction by strategically decreasing its reliance on charcoal while increasing its use of coercive tactics in regions with valuable unsanctioned agricultural commodities. As discussed in the case study section, to compensate for sanction-induced losses, al-Shabaab augmented its predatory behavior in cash crop-producing regions, extorting and killing civilian farmers to obtain these resources, and utilizing forced labor to raise cash crops. These predatory behaviors allowed al-Shabaab to practically render these sanctions ineffective and, in the process, caused widespread human suffering (e.g., U.S. Department of State, 2012; U.N. Security Council, 2016; Radtke & Jo, 2018).

Although targeted sanctions are imposed to achieve their intended aim while minimizing unintended externalities (e.g., Drezner, 2011; Peksen, 2019a), the behavior of al-Shabaab underscores that their impact on armed actors might be counterproductive. Indeed, several scholars argue that the unintended consequences, or second-order effects of sanctions, can have grave implications for human rights and well-being (Biersteker, Eckert, & Tourinho, 2016; Carneiro & Jr, 2016; Early and Schulzke, 2019; Kustra, 2022; Early & Peksen, 2022; Haggard & Noland, 2023). Yet, our understanding of the conditions under which armed actors commit atrocities in response to sanctions – along with their motivations for doing so – is limited (Bapat, Heinrich, Kobayashi, Clifton, & Morgan. , 2013; Grauvogel, Licht, & von Soest, 2017; Hultman & Peksen, 2017; Early and Schulzke, 2019; Allendoerfer, Murdie, & Welch, 2020).

This study combines theory with high-resolution data and qualitative evidence to identify a key implication of targeted sanctions for human rights and security. We focus on targeted sanctions specifically designed to hurt the ability of armed actors to trade in resources, equipment, and assets. As a result, targeted sanctions are designed to cover several key areas, often combining limitations on trade, access to relevant materials, and arms embargos (for instance, in our sample, all sanctioned actors

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* Corresponding author.
E-mail addresses: gurtuzua@iu.edu (J. Urtuzuastigui), okoren@iu.edu (O. Koren).

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that perpetrated violence were targeted simultaneously by sectoral sanctions and arms embargos in addition to resource-limiting sanctions). Facing the resource constraint borne by targeted sanctions, both state and nonstate actors have incentives to engage in resource substitution and compensate for the loss of sanctioned resources by extracting new revenues from unsanctioned agricultural commodities. Cash crops – cotton, coffee, oil, etc. – are almost never sanctioned, considering the potential humanitarian impacts (Early & Peksen, 2022) and the fact that policymakers often believe that cash crops do not foster violence (Markowitz, 2017). As an illustration, Biersteker, Eckert, Tourinho, and Hudáková (2018) dataset records dozens of sanctioned goods and commodities, but (except for timber) not a single sanction is targeted at food or cash crops. We argue that sanctioned actors will gravitate toward locations *where* such cash crops are produced following *times of greater productivity* to appropriate the revenues and benefits (e.g., money from selling crops, looted cash) provided by the greater availability of these resources, and may use violence to facilitate appropriation (Koren & Bagozzi, 2017; Martin-Shields & Stojetz, 2019; Ubilava, Hastings, & Atalay, 2023).

Despite this intuition, little attention has been given to the second-order effects of targeted sanctions on civilian victimization, *especially at the local level*. Researchers study how third parties can impact the decision of armed actors engage in attacks on civilians (Salehyan, Siroky, & Wood, 2014; Tokdemir & Mark, 2018; Asal, Phillips, Karl Rethemeyer, Simonelli, & Young, 2019), but not how the withdrawal of support impacts such behaviors. One complication of studying the latter relates to the possibility that shifts in violence in response to targeted sanctions occur *locally* – the rates of violence at the national level might remain unchanged, but where they happen *within* the country changes. We match information on targeted sanctions from the UN targeted sanctions dataset (Biersteker et al., 2018) with information on one-sided violence events from the Geolocated Event Dataset (GED) (Sundberg & Melander, 2013), identifying – for each GED actor that perpetrated an attack on civilians – whether these attacks were committed by sanctioned actors, pre-sanctioned actors (i.e., sanctioned actors before a sanction was imposed),[2] and unsanctioned actors. We count the total number of attacks for each type of group (sanctioned, pre-sanctioned, and unsanctioned) separately, within the same location – corresponding to the centroid of a 0.5-degree grid cell[3] in the AfroGrid dataset (Schon & Koren, 2022) – and month for each grid cell month between 2006 and 2013.[4] This analysis builds upon prior work that assesses one of the most pernicious second-order impact of targeted sanctions – the targeting of civilians (e.g., Wood, 2008; Peksen, 2009; Hultman & Peksen, 2017). Yet, it is the first study (to our knowledge) that analyzes the impact of sanctions on one-sided violence perpetrated by armed actors specifically as well as to evaluate an inadvertent consequence of sanctions at the subnational and subannual grid-month level.

Directly in line with our theory, the quantitative empirical results reveal a clear pattern whereby sanctioned actors increase the rate of their attacks within cash-crop producing areas following months of greater agricultural productivity compared with other non-cash crop agricultural locations. Moreover, we find that unsanctioned and – importantly, for the purpose of assessing strategic shifts – pre-sanctioned groups do not engage in similar behaviors, and do not increase their attacks on civilians in cash-crop producing areas following months compared with other non-cash crop agricultural cell-months. These results are robust to a battery of sensitivity analyses, which provide greater support to the viability of the hypothesized dynamics.

To ensure that these data trends are driven by the substitution mechanism we hypothesize, we delve deeper into the case of al-Shabaab. Using qualitative and descriptive evidence, we illustrate that the group actively shifted to engaging in looting and violence where non-sanctioned agricultural resources were produced after the UN imposed sanctions on charcoal exports. In line with our theory, we also find that the purpose of violence was to extract revenues and harvested cash-crop produce rather than physically take control over these regions. Crucially, the *seasonality* of the violence also shifted following the sanction to track peak harvest periods. In providing evidence of a systematic relationship at the localized and monthly level, we therefore hope to inform key areas of ongoing research, which focus almost exclusively on sanctions' impact at the *country* level, while assisting policymakers, military strategists, and aid workers to better anticipate the timing and location of victimization by sanctioned actors as well as to respond to such tragedies more effectively.

2. Theoretical argument

2.1. Motivation and concepts

International sanctions are popular policy tools utilized by states and multilateral actors to alter a target's policies, constrain its resources, and/or signal condemnation of a violation of norms. Researchers have studied factors that influence the onset of sanctions (e.g., Kaempfer & Lowenberg, 1988; Cox, Cooper, & Drury. , 2006; Urtuzuastigui, 2023) as well as the conditions under which sanctions achieve their intended goals (e.g., Pape, 1997; Elliott & Hufbauer, 1999; Kelley & Simmons, 2015; Peksen, 2019b; Early and Peksen, 2019). More recent work examined sanctions' political, economic, and humanitarian consequences, highlighting their negative externalities, including increasing the risk of political repression (e.g., Wood, 2008; Carneiro & Cristiane and Laerte Apolinário Jr. , 2016), engendering authoritarianism (e.g., Peksen, Cooper, & Drury. , 2010), and harming social and health services and economic performance (e.g., McLean & Whang, 2021; Haggard & Noland, 2023; Neuenkirch & Neumeier, 2016).

Several scholars specifically examined how sanctions affect the duration and intensity of civil conflicts. Escribà-Folch (2010) argues that, by decreasing uncertainty about the distribution of power, reducing the economic incentives of fighting, and limiting the financial and military feasibility of engaging in warfare, sanctions reduce the length of conflicts and increase the probability of a peaceful resolution. Similarly, Lektzian and Regan (2016) find that sanctions have a particularly salient impact on reducing conflict duration when accompanied by military interventions, which allow for effective enforcement. Radtke and Jo (2018) find that when sanctions constrain and weaken rebels by depressing their income and inducing territorial losses and battlefield defeats, the level of conflict intensity decreases. Hultman and Peksen (2017), however, find that imposed economic sanctions induce governments to escalate their conflicts on the battlefield, although they also find arms embargo reduce country-level risk of conflict intensity by limiting the state's military capacity over time.

Despite these conflict-focused contributions, research has generally ignored how sanctions impact the tendency of both state and nonstate (e.g., rebels, militias, and mercenaries) armed actors to engage in violence against civilians, specifically. Yet, considering that sanctions can pose great strain on resources and – by extension – the durability of armed organizations, targeted actors may strategically shift their behaviors to counter the resulting impact, which can have grave implications for human security and the safety of civilians. In other words, targeted sanctions constitute a *resource shock*, which forces actors to identify and secure alternative sources of revenue unaffected by sanctions. A sourcing strategy shift can take time and involve a significant

[2] In this category we also included actors recorded as previously sanctioned, i.e., recorded violence by these actors after sanctions were lifted. There are very few such instances in our sample, and our results remain statistically and substantially unchanged without including the small number of attacks recorded by groups with this status.

[3] A square of 55km x 55km at the equator.

[4] As we discuss in the empirical section, we limit our analysis to this period to preempt potential endogeneity concerns, while accommodating the availability sanctions data.

use of coercion, considering potential opposition by producers. For instance, building on past work by Hovil and Werker (2005) and Weinstein (2005), Salehyan et al. (2014) organization-level analyses find that the degree of sponsoring an actor receives and the type of principal who provides that support are both directly related to said actor's decision to engage in violence against civilians. Koren and Bagozzi (2017) find that facing sudden food constraints due to climate shocks or intensified conflict induce both state and nonstate actors to attack civilians to facilitate the appropriation of food crops. More generally, Hultman (2007), Hendrix and Young (2014), and Wood (2014) find that armed actors engage in violence against civilians to compensate for scarcities induced by their military weakness. Along with these and other studies (e.g., Azam & Hoeffler, 2002; Butler & Gates, 2012; Fjelde, 2015; Asal et al., 2019; Uzonyi & Demir, 2020), we highlight the importance of resource shocks and constraints in engendering violence against civilians by armed actors, focusing on the role of sanctions therein.

Our working definition of "targeted sanctions" follows Biersteker et al. (2018, 407): "legally binding restrictive measures applied by the UN Security Council (under Chapter VII of the Charter) to coerce targets to change their behavior, constrain them from engaging in proscribed activity, and/or signal a violation of international norms." Because we are particularly interested in the effect of resource constraints on the behavior of warring actors, we focus our analysis on resource-constraining sanctions. Such targeted sanctions often include sectoral, commodity, and financial limitations, as well as individual and organizational restrictions, such as asset freezes and travel bans. This contrasts with comprehensive sanctions, which include total bans on investment and trade between the international community and the target actor(s). Since the 1990s, targeted rather than comprehensive sanctions have been used much more extensively (e.g., Biersteker et al., 2018) to exact "more direct pressure on political elites and their support base while minimizing adverse effects on the general population of a target country" (Peksen, 2019a, 279). Moreover, during same period, the Security Council has also emphasized targeting nonstate actors (rebels, militias, etc.) "to achieve the objective of conflict reduction and ultimately civil war termination" (Radtke & Jo, 2018, 759).

We argue that upon the imposition of resource-constraining sanctions, armed actors are incentivized to seize revenues from valuable non-sanctioned resources to replenish lost income. In so doing, they may shift their behavior in a way that leads to the intensification of violence against civilians to facilitate looting and appropriation in regions where such resources are produced, particularly following times of abundance (e.g., due to sales of produce, cash to pay farm workers, etc.). We argue this shift occurs due to a *substitution logic*, which we explore immediately below.

With respect to sanctions, state and nonstate armed actors fall into one of three categories. The first category covers all *sanctioned* actors. As we discuss below, we expect these actors to shift their violent behavior due to the resource-constraining impact of sanctions. The second category includes actors who are *unsanctioned* and hence do not have an incentive to alter their behavior relative to sanctioned actors. And the third category includes all sanctioned actors in the period *before* being sanctioned, which provides a direct comparison of the same organizations before and during a sanction. As a result, we expect *pre-sanctioned* actors to behave like unsanctioned actors.

2.2. Resource substitution and violence against civilians

During civil war, access to primary commodities is one of the ways armed actors secure the revenues necessary for their operations (Collier & Hoeffler, 2004; Weinstein, 2005; Hovil & Werker, 2005). For instance, Lujala (2009, 52) underscores that a "state with access to large resource revenues may be able to hire more soldiers and equip them better." Bannon and Collier (2003, 30) similarly argue that to "assemble and sustain a fighting force of hundreds or thousands of soldiers, a rebel group needs a regular source of income…[and] successful rebel groups in the developing world…have turned to the natural resource sector." To facilitate the appropriation of revenues and goods to resell within resource rich areas, armed actors may engage in violence. In these situations, violence is used as a means of subjugation (Humphreys & Weinstein, 2006; Wood, 2010), preventing civilian defection (Eck, 2014; Hovil & Werker, 2005), and forcing the civilians to part with valuable resources (Stoop, Verpoorten, & Van Der Windt, 2019; Millán-Quijano & Pulgarín, 2023; Lu & Yamazaki, 2023).

The critical importance of natural resource revenues in supporting armed forces and thus for sustaining civil war therefore directly influences how the Security Council constructs its sanctions regimes. Because targeted sanctions are geared primarily toward ending armed conflicts (Eriksson & Wallensteen, 2015), in using targeted sanctions, the Security Council seeks to "constrain access to resources needed to engage in proscribed activity" (Biersteker et al., 2018, 407) by strategically targeting revenues from natural resources, in addition to other sources of income and technology. As Elliott (2016, 172) emphasizes, senders of sanctions may even "broaden their aim to encompass whole sectors, such as aviation or finance, or commodities, such as oil, diamonds, or timber," with the purpose of blocking a target's access to the international market and economy more generally.

In these regards, the aim of resource sanctions is to reduce the ability of armed actors to obtain revenues related to trade in diamonds, gems, and other natural resources, considering that their ability to profit from these resources "depend on access to the global economy – to its markets, its financial intermediaries, its brokers, its investors, and the foreign companies that often extract a developing country's riches" (Bannon & Collier, 2003, 11). This also means that targeted sanctions drastically reduce access to critical revenue streams from business, trade, and finance, which curtails a target's overall spending capacity (Escribà-Folch, 2010; Drezner, 2015; Early & Cilizoglu, 2020; McLean & Whang, 2021). Unable to legally sell valuable natural resources (e.g., diamonds, timber, petroleum, etc.) or transfer important goods, assets, or financing, sanctioned actors lose access to capacities that would otherwise allow them to continue fighting. For instance, data produced by Biersteker et al. (2018) reveal that sanctions on the export of diamond and timber increased costs for the Liberian government by substantially reducing access to revenue; import restrictions, asset freezes, and diamond sanctions undermined UNITA's military capacity in Angola; diamond sanctions tipped the balance against the Revolutionary United Front in Sierra Leone; and financial asset freezes weakened the Libyan government by restricting access to nearly $50 billion in 2011 (see also Elliott, 2016).

Granted, sanctioned actors may sometimes have an opportunity to evade the impact of sanctions. Rebels usually operate in illicit markets to generate revenues and thus might be well-positioned to sell newly sanctioned goods in the shadow economy. In this case, a crucial determinant is whether an intermediary exists to facilitate trade and find buyers on the global market (Ahram, 2023). While disorganized rebel groups who prey on lootable resources may rely on such intermediaries, organized groups who control territory may forge state certification to engage in ostensibly legitimate business transactions (Sweet, 2021). In other cases, formal political actors residing abroad may receive international recognition and legitimacy (e.g., the U.S.' recognition of the Venezuelan opposition in 2019), which allow them to trade resources in licit markets. And some groups (e.g., Kurdistan Regional Government) may have sufficiently strong standing, allowing them trade with international actors while facing few constraints (Ahram, 2023).

All else being equal, while unsanctioned rebel organizations find ways to illegally operate in international markets, it is more difficult to do so after being targeted with UN sanctions. For instance, after the Security Council imposed a diamond export ban on the RUF – requiring all rough diamonds to be legitimated via the Kimberly Process certification scheme – it was considerably more difficult for the RUF to find international buyers, even though much of the intermediary

infrastructure the group was using to smuggle diamonds outside of the country remind unimpacted. As Biersteker (2014, 134) notes, there was "an almost complete halt to the traffic in illicit diamonds from Sierra Leone to Liberia." Furthermore, even when more organized rebel groups are able to build legal veneers by, for example, coercing bureaucrats to forge state certifications, permits, and licenses, total bans on a critical export from the state itself (e.g., the diamond export ban on Liberia without the option of the Kimberly Process authorization) render these certifications ineffective compared with international standards, as doing business with such domestically-certified groups would still violate international law. Finally, international sanctioning actors can go after intermediaries as well. For instance, since the February 2012 charcoal sanction on al-Shabaab, the U.S. has repeatedly targeted individuals who violate the ban, classifying them – in effect – as terrorists (U.S. Department of Treasury, 2023).

Therefore, while we fully agree that circumventing international sanctions is possible, we also argue that the foregoing constraints make it (far) more difficult for rebels to do business and generate revenues. As these costs mount, sanctioned actors may therefore be compelled to search for new sources of revenue, to substitute – at least partly – for the revenues lost by these constraints.

Another issue relates to the different impacts on state vs. nonstate actors. There are several reasons to believe that sanctioning countries can induce similar revenue losses (see, e.g., Koren & Bagozzi, 2017), even if states have more opportunities to evade their costs. For example, states can redirect trade to third parties who capitalize on lost trade between the sender and target country (Early, 2015). Another is to allow firms to operate in the "shadow economy", for instance, via smuggling goods outside of resource producing areas (Early and Peksen, 2019). Some oil exporting governments, such as Russia and Iran, may be able to circumnavigate the hardest parts of the sanction, although they still face a cost for doing so (Bloomberg, 2023). However, for many less economically powerful countries or non-oil exporters, redirecting lost trade to willing third parties ultimately results in losses for target governments, and allowing businesses to operate in the shadow economy entails high risks and drains state resources. The risks of non-payment and interdiction increase, since contract rights cannot be enforced, and smugglers are often compelled to sell their goods below market value to entice third parties to engage in illicit trade. Even more importantly, from a structural-economic perspective, as Early and Peksen (2019, 823–824) point out, "[b]ecause shadow sector activities cannot be taxed, the growth of shadow sectors will sap government budgets that are already strained by sanctions." Considering these issues, while engaging in resource transfers via alternative informal pathways might be possible, these adjustments come at significant costs even for states, ultimately creating strong incentives for replenishing lost income via alternative means.

In these regards, an important aspect of sanctions is that, unlike other types of resource shocks, their targeted nature means they constitute a shock only to revenues derived from certain types of resources. This offers sanctioned actors an alternative. Instead of trying to compensate for lost revenues from trade in sanctioned resources, they could begin extracting revenues from local commodities that are not covered by the sanctions. Because the sanctions regime is targeted rather than comprehensive, these actors can then benefit from the production of these goods, augmenting their revenue streams in the face of sanctions-induced losses. Engaging in resource substitution therefore provides an optimal sourcing strategy, one that helps replenish lost income from trading in the sanctioned resources. If targeted actors "can obtain access to new income or financial flows, the impact of the sanctions could quickly erode…[allowing targets to] replenish their resources rather quickly" (Elliott, 2016, 176).

One especially valuable set of resources for substitution are agricultural cash crops, including, e.g., fruit, coffee, tea, cotton, and sugarcane. Indeed, except for timber, cash crops have not been covered by targeted sanctions since end of the Cold War, if not since World War II (Biersteker et al., 2018), a consequence of the potential humanitarian implications of sanctioning foodstuffs as well as the belief that cash crops more generally are unlikely to foster violent conflict. As Markowitz (2017) points out, because scholars and policymakers consider agricultural resources to be insufficiently profitable, difficult to access, and monopolized by the state, they are unlikely to finance or drive intrastate conflict, which likely explains why senders neither target these resources nor anticipate the substitution behavior we emphasize in this study.

Yet, in both domestic and international markets, cash crops can fetch sizable profits and, as such, have constituted a sizable portion of armed actors' revenue streams in multiple countries, including Syria, Iraq, India, the Philippines, Thailand, Somalia, and Peru (Crost & Felter, 2020; Jaafar & Woertz, 2016). This has been particularly true over the past several decades, as global demand for cash crops has grown rapidly (Markowitz, 2017). Indeed, between 2001 and 2011, agricultural exports rose "from 400 billion to 1.3 trillion US dollars" (Crost & Felter, 2020, 1489), increasing by 250 percent in least developed countries. Especially following periods of abundance, cash-crop producing areas may offer additional sources of revenues. Farmers living on the property might stash cash on location, having been paid for their labor and for selling crops following harvest. Similarly, these locations may contain harvested goods (e.g., bags of wheat) that could be easily transported to and sold in local or even international markets (e.g., Fjelde, 2015; Jaafar & Woertz, 2016). Accordingly, when armed actors – both state and nonstate – face international sanctions, they have incentives to shift their appropriation efforts into regions where cash crops are produced, looting cash and transferrable harvested crops, thereby effectively smoothing the impact of targeted sanctions on their ability to operate and support themselves.

As a result, as is the case with other types of commodities, such as gems and diamonds (Azam & Hoeffler, 2002; Lujala, 2009; Wood, 2010; Whitaker, Walsh, & Conrad, 2019), in shifting their appropriation efforts to cash-crop producing areas, armed actors may similarly engage in violence against civilians to forcibly extract revenues from civilian producers. The notion that armed actors strategically use violence against civilians to secure income generated from trade in valuable resources during conflict is firmly established in the extant literature (e.g., Wood, 2010; Salehyan et al., 2014). This is especially true when armed actors experience sudden and significant income shocks, as time horizons shorten and thus the more immediate strategic objective of replenishing lost revenues is prioritized over longer-term goals, such as co-opting civilians to obtain critical resources (Wood, 2014; Koren & Bagozzi, 2017; Whitaker et al., 2019; Stoelinga, 2023). Although such violent tactics are more likely to entail both local and international reputational costs – weakening a target's ability to establish or maintain economic support over the long-term – sanction-induced income shocks mean that the targeted armed actor is less able to provide "positive incentives [and thus] must seek alternative short-term solutions," particularly "increases in victimization in the communities that produce the commodities" (Estancona, 2021, 1114).

It is important to emphasize that these tactics are used in cash crop regions despite being labor intensive, as many of these regions are highly centralized, which lowers the transaction costs of violent extortion against farmers and exporters, making it a lucrative source of income. The incentives to prey on export crops are particularly strong, as "they are usually marketed through highly concentrated value chains with a small number of 'choke points' that can easily be held up by armed groups" (Crost & Felter, 2020, 1485). For example, rebels and militias generate large revenues from extorting banana and sugar firms in the Philippines and avocado businesses in Mexico, violently attacking plantations and processing plants when businesses refuse to make extortion payments (e.g., Ornelas, 2018; Crost & Felter, 2020; Herrera & Martinez-Alvarez, 2023). In more decentralized cash crop regions, however, such as in Bungudu, Nigeria, armed actors are known to criminally extort small farmers and farming-dependent villages,

kidnapping, torturing, and murdering civilians to punish non-compliance or in response to civilian vigilantism (Buba, 2023). Similarly, in the Central African Republic, "the coffee trade, harvest cycles, and control of key markets" have contributed to a significant increase in violence against farmers within the country's coffee regions (Mangan, Acko, & Taha, 2020, 1). And, as we show in detail in our short case study, al-Shabaab's behavior in Somalia also illustrates the use of violence for the purpose of extorting cash crops and the resulting revenues from civilians.

Notably, the revenue-generating potential of cash crops partly explains why there are "higher incidences of local violence in weak states with cash crop economies than in weak states with mobile, non-renewable resources" (Markowitz, 2017, 345). The surpluses and revenues generated by valuable agricultural resources may therefore lead armed actors to violently extort farmers or large export firms (Crost & Felter, 2020), which intensifies civilian victimization "associated with looting, forced labor, or the expulsion of local communities" (Addison, Le Billon, Mansoob, & Murshed, 2002, 370). Crucially, because cash crops tend to be produced and sold seasonally, the strategic use of violence to acquire these resources likely intensifies during specific months during the year where produce is likely to be high (Ubilava et al., 2023).

As a result, this type of resource-substitution driven violence will not necessarily be observable in analyses that focus on the country level – the rates of violence against civilians nationally may remain unchanged, but the *location* where such attacks intensify will shift. Yet, as mentioned above, research on the impact of sanctions generally left their local-level effects, with respect to civilian victimization or otherwise, unexplored. Additionally, while few studies have assessed the relationship between crop productivity and violence at the local level (Crost & Felter, 2020; Koren & Bagozzi, 2017), in addition to never examining targeted sanctions' impacts on violence, by and large, these studies also employ annual data on crops and violence. This means they cannot clearly identify the impact of *intra*-annual variations and the more exact timing of armed actors' strategic response to stress, even though such variations likely have much greater effect on violence compared with *inter*-annual ones (Koren & Schon, 2023). Therefore, an effective measure of the substitution effect we describe above should capture not only the difference across cash crops and non-cash crops within agricultural locations, but also whether violence changes following months when production is higher versus times when it is lower within these areas.

Building on our theoretical discussion, we hence expect that in cash-crop producing areas, resource substitution incentives will induce sanctioned actors to increase the number of their attacks on civilians, but only following months where such resources are available, compared with other agricultural areas that produce non-cash crop (e.g., vegetables) and other valuable resources (e.g., diamonds). At the same time, we do not expect the rates of violence by unsanctioned actors and pre-sanctioned actors to follow the same incentives, seeing that they are not facing pressures to shift to cash-crop appropriation. These logics suggest the following hypotheses:

Hypothesis H1: *Sanctioned actors will increase their violence against civilians rates in cash-crop producing locations following months of greater agricultural productivity compared with other agricultural locations.*

Hypothesis H2: *Violence against civilians rates by unsanctioned and pre-sanctioned actors will be unaffected by cash crop and cash crop productivity compared with other agricultural locations.*

Note that even if sanctioned actors do choose to rely on a shadow economy or smuggling to compensate for the impact of sanctions, these behaviors should bias any estimates *downward*, that is, away from a positive sign and statistical significance. As such, H1 is a stringent test of our hypothesis, namely that sanctions increase the risk of appropriative violence and looting of revenues even if actors also try to simultaneously circumnavigate the sanctions. Any estimates in case of a confirmation (positive relationship) can hence be considered, if anything, conservative, seeing that this suggests the pressures we hypothesize work despite of the existence of a shadow economy.

3. Quantitative analysis

3.1. Sample, variables and identification

We test our hypotheses using a spatially and temporally disaggregated analysis of AfroGrid, a recently released monthly dataset incorporating a wide variety of conflict, development, and climate variables measured at the 0.5-degree cell (~55 km × ~ 55 km at the equator) month level (Schon & Koren, 2022). Our sample is constrained to the Jan. 2006 – Dec. 2013 period due to the empirical availability of our key dependent and independent variables. While the focus on Africa might somewhat limit our inference, it is useful insofar as this region has experienced most ongoing domestic conflicts during our period of analysis (Adams et al. 2018). Additionally, as mentioned above, we are interested in contrasting violence in cash-crop producing to violence in other (non-cash crop) agricultural areas. To this end, we follow Baogzzi, Koren, and Mukherjee (2017) and retain only agriculturally producing cells according to the Spatial Production Allocation Model dataset (SPAM) (Yu et al., 2020). There are 6,460 (cash and non-cash) crop-producing agricultural grid cells in this sample (out of a total of African 10,745 grid cells), which are measured over the span of 96 months, for a total number of 620,160 cell-month observations.[5]

We operationalize our dependent variables measuring violence against civilians (VAC) by sanctioned, unsanctioned, and pre-sanctioned actors for each cell month in our sample in three steps. We begin by using data from the Uppsala Conflict Data Program's Geolocated Event Dataset (GED) (Sundberg & Melander, 2013), which records information on state-based and nonstate-based conflicts as well as one-sided violence against civilians events involving at least one death and occurring as part of an ongoing war.[6] We specifically retain only incidents of one-sided violence, which we refer to hereon as violence against civilians. In the next step, we rely on monthly data on targeted sanctions imposed by the UN Security Council on state and nonstate (rebel and militia) actors (Biersteker et al., 2018) to code, for each actor in the GED, whether they were under a UN sanction in a given month. Having created a dataset of all VAC events by sanctioned actors, we then aggregate them to the cell month level, while creating separate indicators for VAC attacks by unsanctioned and pre-sanctioned (i.e., the same sanctioned actors *when they did not face active sanctions*) actors.[7] The variables *Sanctioned* VAC_{it}, *Unsanctioned* VAC_{it}, and *Pre-sanctioned* VAC_{it} have means 0.001, 0.001, and 0.0002 and ranges $0 \Leftrightarrow 26$, $0 \Leftrightarrow 20$, and $0 \Leftrightarrow 11$, respectively, over the Jan. 2006 – Dec. 2013 period. To mitigate the risk of outliers' impact, we log these variables (and their lags) before entering them into our model, while ensuring the results are robust both to our decision to log these variables and to their count model nature, in the Supplemental Appendix.

Our key independent variables must approximate the impact of (greater) cash crop productivity on violence. To construct it, we follow

[5] This also ensures we are maintaining a higher threshold of robustness for any statistical effects.

[6] Defined as a conflict with at least 25 combatant casualties.

[7] Another way of testing the effects of sanctions is to create a grid-month-actor data framework, where every actor is recorded in every agricultural grid cell for every month between Jan. 2006 and Dec. 2013, and then assessing whether the behavior of these actors shifted as the sanctions – now an independent variable as part of a triple interaction with cash crops and productivity – are deployed. Doing so, however, will increase the size of the sample by an order of magnitude that cannot be feasible done or analyzed using the computational resources available to us. The separation of sanctioned and pre-sanctioned actors on the dependent variable allows us to conduct exactly the same comparison without the need to expand our data frame to these magnitudes, while also allowing us to assess a more intuitive simple (twoway) rather than a threeway interaction.

Koren and Schon (2023) approach. We first identify all locations where cash crops are produced based on data from the Spatial Production Allocation Model dataset (SPAM) (Yu et al., 2020). SPAM identifies a variety of crops produced locally across the globe for the years 2000, 2005, 2010, and 2017, recoding the 42 primary crops produced annually in each 0.5 AfroGrid cell. To construct our *Cash crops*$_i$ indicator, we then identify all locations where cash crops – defined as oil crops (e.g., sunflower, rapeseed), bananas, plantains and other fruit, cotton, coffee, and sugarcane, in line with past research (Crost & Felter, 2020; Koren & Schon, 2023) – are produced (=1), distinguishing them from other agricultural grid cells, where no cash crops were produced (=0). To ensure that any variability in productivity of cash crops is not due to potential reallocation of regions in the sample due to violence by armed groups, we rely on the 2005 SPAM map values, analyzing violence starting in January 2006, while ensuring in the Supplemental Appendix that our results are robust to this decision.

Having identified cash crop locations, we then create a variable of productivity for each cell-month in our sample (Koren & Schon, 2023). We rely on the Normalized Difference Vegetation Index (NDVI), a continuous indicator of vegetation and agricultural productivity, which ranges from 0 (no vegetation) to 1 (the entirety of the cell-month is covered by vegetation) on land, although negative values are possible in some cases (e.g., lake shores) where the cell's surface has slightly more water than land. NDVI information was obtained from the MODIS Terra monthly satellite data, downloaded and processed with the MODIStsp R package (Busetto & Ranghetti, 2016) and included in AfroGrid. We lag this variable to account for the time that available produce might take to be appropriated while accounting for this decision in the Supplemental Appendix. *Productivity*$_{it-1}$ is therefore an intra-annual indicator of general "greenery" levels in the previous month within each grid cell in our sample.

Finally, to capture the impact of cash crop productivity, the hypothesized driver, we create the interaction term *Cash crops*$_i$ *X Productivity*$_{it-1}$. This interaction allows us to specifically identify variations in agricultural productivity in the previous month within cash crop producing cells and how they affect by each actor type of interest. Importantly, at the same time, we can also distinguish these impacts from the effect of non-cash crop producing areas as well from the effect of vegetation health generally.[8] For illustration, Figure A.1, Supplemental Appendix, plots maps summarizing the total rates of violence by each type of actor, in addition to a map of all cash crop producing locations.

In addition to these explanatory variables, our models also add variables to account for confounders and alternative explanations, including: (1) whether a given location included secondary diamonds, gems, or gold (Tollefsen, Ha ̈vardStrandand, & Halvard, 2012); (2) local annual population densities, for instance, to account for the possibility that VAC by each actor increases due to the number of more available "targets;" (3) annual total levels of nighttime light emissions, to account for the impact of local development and state capacity levels; and (4) (ln) temperature and (5) (ln) precipitation, to account for the impact of climate on agriculture (all four controls are included in AfroGrid). In line with econometric research recommendations (Angrist & Pischke, 2009), we also include (6) a one-month lag of each respective dependent variable to account for VAC history's impact as a predictor, in addition to lagging all time-varying indicators by one month. To account for all country level factors that might confound our local effects, our baseline models include fixed effects by country and month, and our full models include fixed effects by country X year. Summary statistics of all variables used in all main and robustness models are reported in Table A.1, Supplemental Appendix.

[8] Considering, as we show below, that none of the interactions has a negative and statistically significant coefficient, endogeneity is not an overriding empirical concern, considering any such biases will push estimates downward.

In line with standard econometric approaches (Angrist & Pischke, 2009), we estimate a series of fixed effects linear models estimated using ordinary least squares (OLS) regression, while accounting for the (zero-inflated) count nature of our dependent variables in the Supplemental Appendix, relying on two general identification formulas[9]:

$$\ln y_{it} = \beta_1 \boldsymbol{c}_1 + \beta_2 \boldsymbol{n}_{it-1} + \beta_3 \boldsymbol{c}_i \times \boldsymbol{n}_{it-1} + \beta_4 \boldsymbol{r}_i + \beta_{5-6} \ln \boldsymbol{X}_{it-1} \beta_{7-8} \ln \boldsymbol{S}_{it-1} + \beta_9 \ln \boldsymbol{y}_{it-1} + \omega_j + \phi_t + \epsilon_i \quad (1)$$

$$\ln y_{it} = \beta_1 \boldsymbol{c}_1 + \beta_2 \boldsymbol{n}_{it-1} + \beta_3 \boldsymbol{c}_i \times \boldsymbol{n}_{it-1} + \beta_4 + \beta_{5-6} \ln \boldsymbol{X}_{it-1} \beta_{7-8} \ln \boldsymbol{S}_{it-1} + \beta_9 \ln \boldsymbol{y}_{it-1} + \omega_j + \phi_t + \omega_j \times \psi_T \epsilon_i \quad (2)$$

In Equation (1), ln $\mathbf{y_{it}}$ is a vector of (ln) VAC event counts by sanctioned, unsanctioned, and pre-sanctioned actors, respectively, and $\mathbf{y_{it-1}}$ their corresponding one-month lags; $\mathbf{c_i}$ is an indicator of whether cash crops were grown in a given grid cell, $\mathbf{n_{it-1}}$ is a vector capturing vegetation productivity the previous month, and $\mathbf{c_i}$ x $\mathbf{n_{it-1}}$ their interaction measuring cash crop productivity; $\mathbf{r_i}$ is an indicator of valuable non-agricultural resources in each grid cell; $\mathbf{X_{it-1}}$ is a matrix of our (logged) socioeconomic controls (nighttime light emissions and local population); $\mathbf{S_{it-1}}$ is a matrix of our (logged) climatic controls (rainfall and temperature); ω_j and ϕ_t are fixed effects by country and month, respectively[10]; and ε_i are standard errors clustered by grid cell. In Equation (2), we then add fixed effects for year (ψ_T) and their interaction with country fixed effects to account for all annual (in addition to constant, as in Equation (1) country level features.

4. Results

Table 1 first reports a baseline model (Equation (1) of VAC by sanctioned, unsanctioned, and pre-sanctioned actors, which includes our key natural resource and socioeconomic controls in addition to country and month fixed effects.[11] This baseline model is followed by the full specification (Equation (2), which adds country X year fixed effects to account for all time-varying country level confounders. All results strongly support the hypothesized effect of higher cash crop productivity on VAC by sanctioned actors, but not by unsanctioned or pre-sanctioned ones. Most importantly, the coefficient of our interaction term, *Cash crops*$_i$ X *Productivity*$_{it-1}$, is positive and statistically significant (to the $p < 0.05$ level) in both sanctioned actor VAC models, but insignificant and practically nil in the unsanctioned and – importantly – pre-sanctioned VAC models. In line with our hypotheses, this suggests that the rates of VAC by sanctioned actors increase in cash crop producing location during months of high productivity, which is not the case for other actor types, although we must visualize these effects to ascertain whether they are substantive.

Examining the constitutive terms of the interaction, across all actor types, it appears that following months of no productivity, cash crop producing location are expected to experience VAC by at a lower-than-average rate, a relationship that is only statistically significant for sanctioned actors, who seek resources to appropriate. Additionally, in non-cash crop agricultural grid cells, productivity has a negative but statistically insignificant impact on VAC by all actors. Overall, then, Table 1 confirms both hypotheses H1 and H2, and suggests that only VAC by sanctioned actors is sensitive to cash crop productivity. We also

[9] Where *i* corresponds to the agricultural grid cell, *j* to the country, and *t* to the month.

[10] The time-invariant nature of *Cash crop*$_i$ and *Profitable resources*$_i$ means that we are unable to include grid cell fixed effects in our models without having these indicators omitted from analysis.

[11] A minimalist specification that includes only our key explanatory variables and their interaction alongside country and month fixed effects is reported in Table A.2, Supplemental Appendix.

Table 1
Determinants of Violence Against Civilians in African Locations, Jan. 2006– Dec.2013.

	Baseline				Full			
	Sanctioned (1)	*Unsanctioned* (2)	*Pre-sanctioned* (3)	*Sanctioned* (4)	*Unsanctioned* (5)	*Pre-sanctioned* (6)
Cash $crops_i$	−0.001*	−0.0001	0.00000	−0.001*	−0.0001	0.00000
	(0.0004)	(0.0001)	(0.0001)	(0.0004)	(0.0002)	(0.0001)
$Productivity_{it-1}$	−0.0004	−0.0002	−0.0003	−0.0004	−0.0003	−0.0004*
	(0.001)	(0.0004)	(0.0002)	(0.001)	(0.0004)	(0.0002)
Cash $crops_i$ X $Productivity_{it-1}$	0.002**	−3.905e-06	0.0001	0.002**	−1.201e-06	0.0001
	(0.001)	(0.0003)	(0.0003)	(0.001)	(0.0003)	(0.0003)
Profitable resourcesi	−0.0005**	−0.0003	−0.0001***	−0.0005**	−0.0003	−0.0001***
	(0.0002)	(0.0003)	(0.0001)	(0.0002)	(0.0003)	(0.0001)
DV_{it-1}	0.327***	0.358***	0.221***	0.325***	0.355***	0.219***
	(0.055)	(0.122)	(0.065)	(0.055)	(0.122)	(0.065)
Ln $Population_{it-1}$	0.0003***	0.0001***	0.0001***	0.0003***	0.0001***	0.0001***
	(0.0001)	(0.00003)	(0.00003)	(0.0001)	(0.00003)	(0.00003)
Ln NTL_{it-1}	0.0001	0.0001***	−0.00001	0.0001	0.0001***	−0.00001
	(0.0001)	(0.00002)	(0.00001)	(0.0001)	(0.00002)	(0.00001)
LN $Temperature_{it-1}$	−0.001*	−0.0003	−0.0003**	−0.001*	−0.0003	−0.0003**
	(0.0005)	(0.0003)	(0.0001)	(0.0005)	(0.0003)	(0.0001)
LN $Precipitation_{it-1}$	−0.00000	−0.00004**	0.00003**	−0.00000	−0.00004*	0.00003**
	(0.00003)	(0.00002)	(0.00001)	(0.00003)	(0.00002)	(0.00001)
Observations	591,921					
R^2	0.110	0.123	0.050	0.110	0.125	0.051
$Adjusted R^2$	0.110	0.123	0.050	0.110	0.125	0.051

Standard errors clustered on grid cell in parentheses; Ln corresponds to Natural Log; fixed effects by month and country were included in each regression, although none is reported here.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

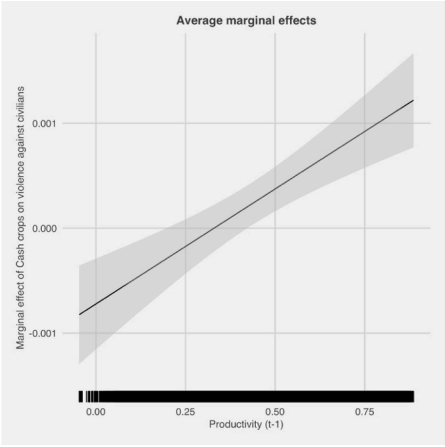
observe that areas with non-agricultural valuable resources (diamonds, gems, gold) are negatively associated with lower VAC rates by all actors, which –in the sanctioned actors' case – is in line with the possibility of resource substitution. Overall, the results in Table 1 provides strong support for our theoretical expectations.

To substantively evaluate the impact of our interaction term in each actor category, we use our full model estimates for each dependent variable to calculate the change in *Cash* $crops_i$'s coefficient on the expected (log) number of VAC across the range of $Productivity_{it-1}$ ($-0.14 \Leftrightarrow 0.89$). We plot these estimated marginal effects for our three full models (rescaling the y-axis in each plot to match the sanctioned actor VAC plot, for illustration), along with their 95 % confidence intervals, in Figure 1. In Figure 2 we compare the effect of productivity across agricultural cells with low (=0) and high (=1) and cash crop value. Examining the sanctioned VAC plot (top), we can first observe that the effect of *Cash* $crops_i$ on violence where there is no productivity is negative and different from zero – cash crop areas during these months are expected to experience a *reduction* of 0.0005 in the number of (ln) VAC attacks by sanctioned actors, or (considering the average ln number of such attacks in the sample is 0.0006 per cell month) a reduction of about 82 % from the baseline (Figure 1), and that productivity has no impact on violence (Figure 2). However, as $Productivity_{it-1}$ rises, the coefficient of *Cash* $crops_i$ also becomes larger, such that during high productivity months the same cells are expected to experience about 0.0012 *increase* in ln VAC events by sanctioned actors, on average, or about 197 % increase from the baseline (Figure 1), with higher (lagged) productivity now exerting a strong impact on violence.

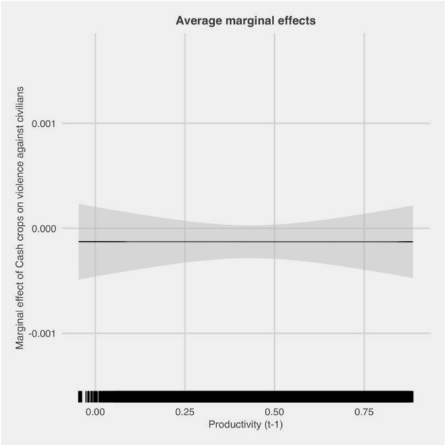
For comparison, we turn to examine the substantive effect of the same *Cash* $crops_i$ X $Productivity_{it-1}$ interaction on VAC by unsanctioned and pre-sanctioned groups. The average effect of *Cash* $crops_i$ is practically constant at about 0.0002 in the number of log VAC attacks by unsanctioned actors (bottom-left) and negative, meaning that regardless of productivity levels, attacks by unsanctioned actors are, on average, less likely in cash-crop producing locations compared with other agricultural cells. Even more interestingly, the rates of VAC attacks by *pre-sanctioned* actors – the same actors in the sanctioned actor category before they were sanctioned (bottom-right) – are nearly identical to unsanctioned actors' violence rates and are markedly different from the level of violence by sanctioned actors: VAC rates by pre-sanctioned actors are unimpacted by changes in productivity and remain zero across the entire range. In both cases, Figure 2 shows that lagged productivity has little-to-no impact on violence by unsanctioned and pre-sanctioned actors. The lack of substantive impact of cash crop productivity on VAC rates by both unsanctioned and pre-sanctioned actors suggests they do not engage in similar substitution behaviors, thereby lending additional support to our theory's viability.

4.1. Sensitivity analyses

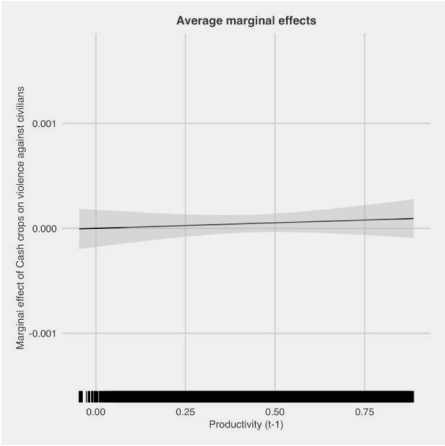
To test the sensitivity of our results to data choices and confounders, we estimate and discuss a battery of robustness models, which – in most cases – correspond to our full model specifications in the Supplemental Appendix. Briefly, we test our results' sensitivity by: (1) estimating a baseline model with only our key explanatory variables and interaction (Table A.2); (2) not lagging our independent variables (Table A.3); (3) using time varying crop indicators (Table A.4); (4) not logging our dependent variables (Table A.5); (5) accounting for the number of civil war events on violence (Table A.6); (5) limiting our sample only to countries that experienced civil war (table A.7); (6) using a zero-inflated negative binomial model (Table A.8); (7) distinguishing between state and nonstate sanctioned actors (Table A.9); (8) using civilians casualties to operationalize sanctioned actor violence rates (Table A.10); and (9) using sanctions as an independent variable in simplified actor-grid-month framework (Table A.11). Crucially our findings hold across in every case, although note that in Table A.9, while the *Cash* $crops_i$ X $Productivity_{it-1}$ interaction is positive in both cases, it falls short of



(a) Sanctioned actors



(b) Unsanctioned actors



(c) Pre-sanctioned actors

Figure 1. The Marginal effect of *Cash Crops*$_i$'s coefficient across the range of *Productivity*$_{it\text{-}1}$.

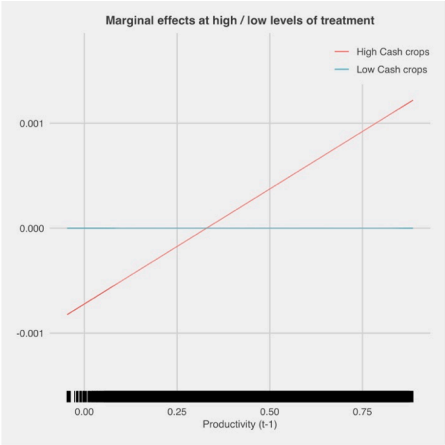
significance for VAC by sanctioned state actors (p = 0.366). As mentioned above, one explanation is the fact that state actors tend to generally be far less likely to be sanctioned than nonstate actors: the number of cell-months in our sample that experienced at least one VAC attack by sanctioned state actors was 196, compared with 552 cell months that experienced at least one attack by sanctioned nonstate actors. Nevertheless, that the interaction's coefficient maintains its sign does suggest that our theory is viable for state forces.

Finally, expecting (per our theory) that the more effective a sanction is in achieving its intended aim (to constrain armed actor resources), the greater the impact this will have on the rates of civilian victimization in cash-crop producing areas following productive months, we follow Biersteker et al. (2018) definitions to compare the impact of "partly effective" and "highly effective" sanctions. The model estimates are reported in Table A.12, and *Cash crops*$_i$ X *Productivity*$_{it-1}$ interaction coefficient for each sanction case is visualized in Figure 3, which shows that highly effective sanctions have a larger impact on sanctioned VAC rates. In other words, the more effective the targeted sanction is in blocking armed actors' revenue streams, the more they will be incentivized to seek alternative cash-crop resources, using violence to facilitate appropriation, as stipulated by our theory.

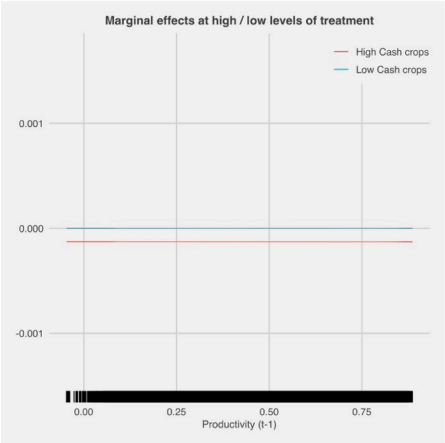
5. Validating the mechanisms using qualitative evidence from Somalia

To establish the viability of the underlying mechanism we hypothesize, we additionally conduct a targeted case study, selected using Seawright and Gerring (2008) "typical" case selection approach. We seek a case where we can assess the impact of a specific targeted resource sanction on violence over space and time, distinguishing it from the history of violence prior to the sanction's imposition. We therefore focus on Harakat al-Shabaab al-Mujahideen, al-Shabaab for short. While the exact date is unclear, al-Shabaab was established as rebel group in 2008 (Radtke & Jo, 2018). Since its inception, the group has been notorious for engaging in both armed conflict and especially attacks on civilians, with terrorist attacks that included car bombs and massacres, which has claimed thousands of lives. Since its inception, the group has shown great flexibility in being able to secure funding to support its operations. As Levy and Yusuf (2021, 1167–1168) note: "al-Shabaab originally relied on donations from diaspora communities and charcoal exports, but became limited in its ability to collect external contributions as a growing global effort to thwart terrorist funding increasingly closed international funding routes. In response, al-Shabaab has created internal revenue sources through taxation, port fees, racketeering, and possibly piracy." Similarly, Radtke and Jo (2018, 780) find that "the Somali-based group's survival was a function of its ability to quickly adapt and divert economic resources in response to changing circumstances." They also find that, for the group, "the key sanction episode was the 2012 charcoal ban" (Ibid.).

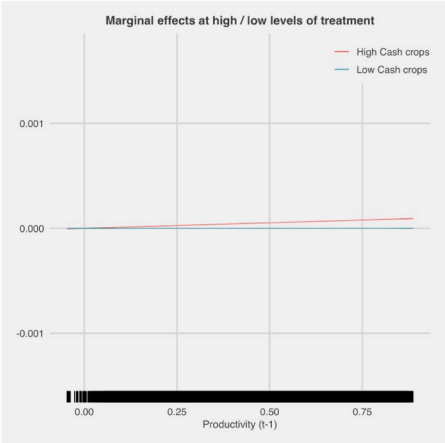
What impact did this ban carry on the group's resource base and revenue streams? Al-Shabaab has circumvented the costs of charcoal sanctions by establishing an efficient extortion racket that targets a wide



(a) Sanctioned actors



(b) Unsanctioned actors



(c) Pre-sanctioned actors

Figure 2. The Marginal effect of *Productivity*$_{it-1}$ across high/low cash crop levels.

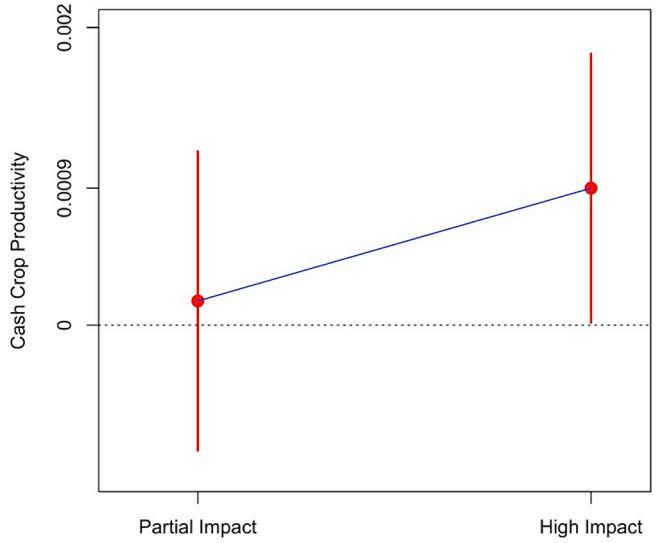


Figure 3. Change in *Cash crops*$_i$ X *Productivity*$_{it-1}$ coefficient on *sanctioned VAC*$_{it}$ based on sanction efficacy.

variety of agricultural produce, including maize, sorghum, potatoes, squash, and many other cash crops, murdering or otherwise using extreme violence against those who fail to comply with its demands (Bahadur, 2022). Moreover, according to UN monitoring groups, "Al-Shabaab has also proved adept at offsetting declining charcoal revenue by the increased taxation of other natural resources and commodities, such as the illicit sugar trade, agricultural production in southern Somalia and livestock in central Somalia" (U.N. Security Council, 2016).

That al-Shabaab increased its reliance on unsanctioned agricultural goods and extortion of farmers in rural areas is directly in line with the substitution logic we hypothesize. To further evaluate this, we leverage primary media reports from the period around the imposition of the charcoal ban (2012–2014) to identify how the group has shifted its non-sanctioned goods related appropriation activities. One example of such an agricultural good is khat, a leafy plant that is chewed by many in the Horn of Africa due to its mild narcotic and stimulating properties. Unlike, say, coca, growing and importing khat is fully legal, not only in Somalia and Kenya, but also in most European states. Because production and export are legal, it offers a useful substitute to banned products, seeing that "[t]here are legitimate businesses whose end product may end up in the hands of Al Shabaab" (McKenzie & Teo Kermeliotis, 2012). The problem – or advantage from al-Shabaab's perspective – is that considering many producers depend on the plant, figuring out whether and how to ban its import "is a precarious balancing act - governments need to think of mechanisms that ensure that these legitimate businesses

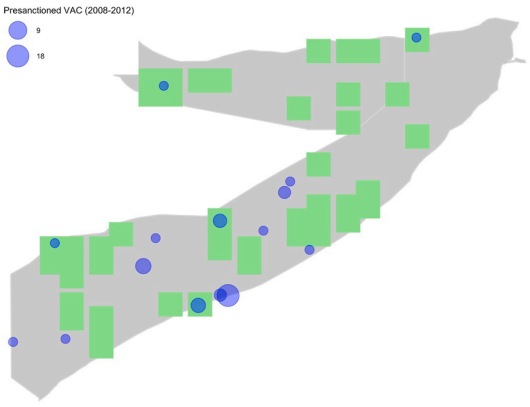
don't end up benefiting these terrorist organizations" (McKenzie & Teo Kermeliotis, 2012). Al-Shabaab could hence force farmers to grow khat – which is legitimately sold – then appropriate the revenues.

A second mechanism that allowed al-Shabaab to compensate for the ban was through extorting farmers in rural agricultural land. The charcoal ban imposition coincided to some extant with a counterattack by Kenyan forces that pushed al-Shabaab out of some major cities in Somalia and into the countryside (U.S. Department of State, 2013; Barasa, 2014). In these rural contexts, al-Shabaab had strong incentives to appropriate – using violence – farmer revenues from agricultural production. For instance, in Somalia's Lower Shabelle region, "al-Shabaab had imposed hefty taxes on residents' crops, seized portions of their crop yields and threatened farmers who said they could not pay" (Jibril, 2013). Such "taxation" measures forced farmers to "pay $300 each month" (Jibril, 2013), a substantial sum in these rural areas, which is "equal to about 40 to 50 per cent of their income" (Presse, 2012). As one farmer notes, "[t]heir [al-Shabaab's] fighters used to keep a close eye on our farms as though they had invested in them" (Jibril, 2013). Farm produce wasn't only used as a source of revenue. As the same farmer added, "Al-Shabaab used the crops it had commandeered to feed its fighters" (Jibril, 2013). Farmers who would not submit to al-Shabaab's demands, would be punished. As another farmer noted, if one refused to pay the rebels, "they would either label you as a heretic or their fighters might even kill you in cold blood" (Presse, 2012).

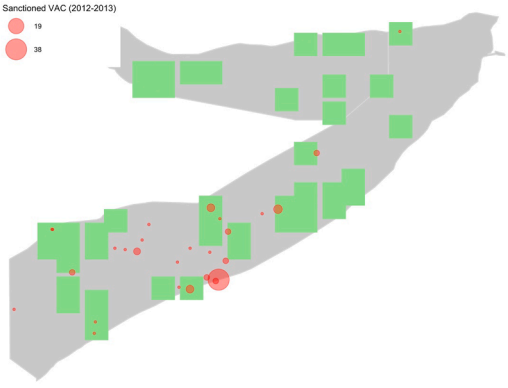
Such violence was not unique to areas under the group's control. In other locations, al-Shabaab fighters would raid farms, looting money and even harvesting crops. According to a farmer, in one incident, "the gunmen went on a shooting spree, forcing villagers to run for their lives into the forest on the outskirts of the village" (Boniface, 2014). The same farmer added that "then the gunmen came right to the farm, harvesting the maize for at least two hours…At daybreak I took stock of the haphazard harvest and they had covered about two acres of my farm" (Ibid.). This evidence suggests that rebels may not only seek to loot revenues from produce or previously harvested crops; they may even spend time to harvesting unsanctioned agricultural produce independently for the purpose of feeding their troops or selling it on local markets.

To visualize the impact of the February 2012 charcoal ban on the rates and locations of al-Shabaab's violence, we summarize and visualize GED, productivity, and SPAM data in the maps and figures below in Figure 4. The top plot sums the total al-Shabaab attack rates reported in the GED between February 2008 and (including) January 2012, the month before the imposition of the charcoal ban. The center plot sums al-Shabaab's attacks for the 20 months after the imposition of the ban (February 2012 – December 2013). The bottom plot does the same for the four years after the ban's imposition (February 2012 – December 2016). In each plot, cash-crop producing areas (according to SPAM's 2005 data) are plotted as light/green squares. These visualizations suggest some interesting trends over time. First, the rates of al-Shabaab's attacks on civilians the 20 months after the ban's imposition are higher than the rates of violence in the four years preceding it, as are the rates of attacks the four years after the sanction's imposition. Second, the locations where the group perpetrated violence also shifted. Between February 2008 and January 2012, only four cash-crop producing locations experienced violence by al-Shabaab, while in the 20 months following the charcoal ban, 11 cash-crop producing locations experienced such violence; and between February 2012 and December 2016, 20 cash-crop producing locations experienced al-Shabaab violence. It appears the group may have shifted its strategic sourcing approaches following the ban, although as mentioned above, some of this violence may also be attributed to the al-Shabaab's response to Kenya's counterattacks (U.S. Department of State, 2013).

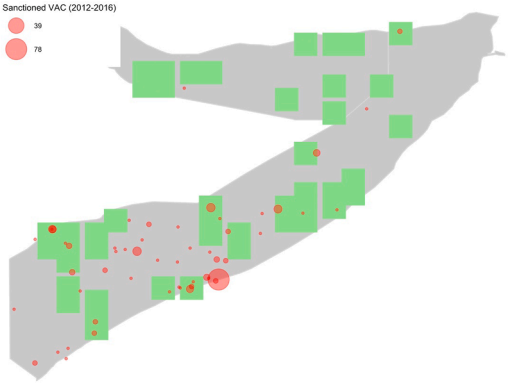
If the charcoal impact ban impacted al-Shabaab according to the hypothesized mechanism, then another expectation is that is that their monthly violence pattern would shift following the ban to follow productivity patterns more closely, such that violence is the highest



(a) Pre-sanction violence against civilians by al-Shabaab (February 2008 – January 2012)



(c) Post-sanction violence against civilians by al-Shabaab (February 2012 – December 2013)

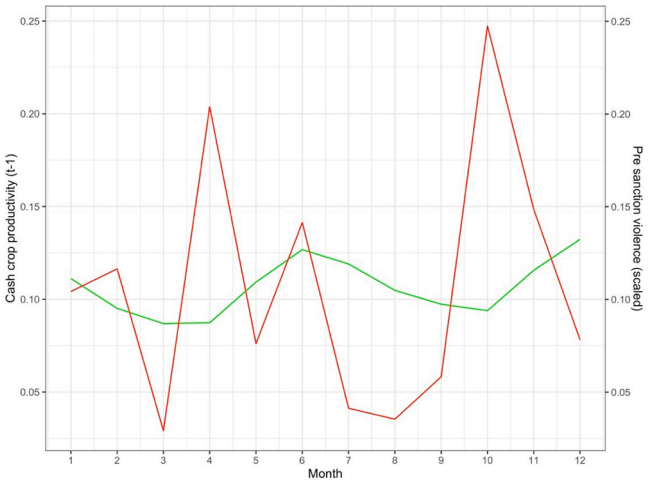


(c) Post-sanction violence against civilians by al-Shabaab (February 2012 – December 2016)

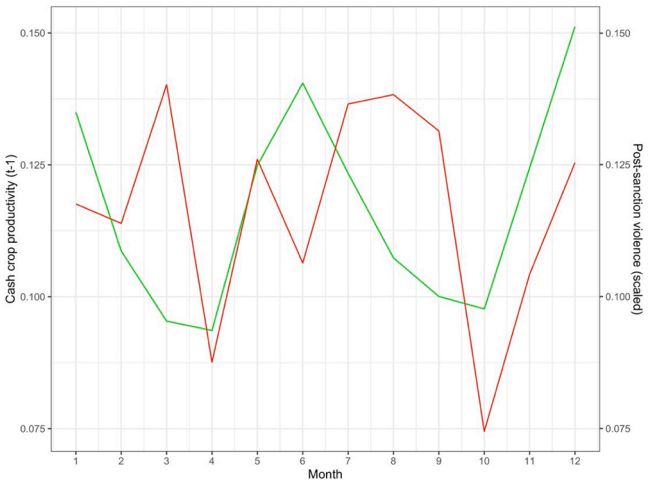
Figure 4. Change in the number of violence against civilians attacks by al-Shabaab after charcoal sanctions (cash cropland in bright color).

following months of peak productivity, which correspond to the availability of pre- or post-harvest goods and cash. Van Weezel (2016) and Koren and Schon (2023), for example, conduct monthly analyses and observe that growing seasons attract more violence, a finding which yearly time scale analyses fail to identify. In line with this logic, Figure 5 plots monthly averages in $Productivity_{it-1}$ within cash crop cells (green/light) and violence against civilians (one sided violence from the GED), scaled to correspond to the relative sizes of variations in productivity. The top plot reports monthly averages for 2008–2011, that is, before the charcoal ban, and the bottom plot reports monthly averages for 2012–2016, that is, after the ban's imposition.

Examining the pre-sanctioned period (top plot), no observable



(a) Pre-sanction (four-year average)



(b) Post-sanction (four-year average)

Figure 5. Monthly cash-crop productivity (green/light) and l-Shabaab VAC (red/dark) averages. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

relationship between (one-month lag) higher cash crop productivity and al-Shabaab violence is apparent; violence rates rise and fall independently of productivity. However, the relationship observably changes during the post-sanction period. Here, we can clearly observe peaks in violence following periods of high cash crop productivity. Overall, Figure 5 suggests that the monthly violence patterns of al-Shabaab changed due to the charcoal ban, and that the shift directly corresponds to variations in cash crop productivity, which – along with the qualitative and visual evidence presented previously – is in line with our hypothesized substitution mechanism, and our theory broadly.

6. Scope conditions and alternative explanations

Considering our analysis if focused on Africa, the first scope condition suggests that similar resource substitution patterns might not be applicable to other regions. Nevertheless, the literature demonstrates that resource shocks often lead to civilian victimization beyond Africa, and that cash crops are susceptible to looting and violent extortion in conflict-prone countries across the Middle East, Asia, and the Americas (Crost & Felter, 2020; Jaafar & Woertz, 2016). Yet, more research is needed to confirm that the sanction-centric trends and mechanism we identify in African states are valid on a global scale.

A second scope condition relates to situations where actors might benefit from being sanctioned. For instance, Woertz (2019) shows that during the multilateral United Nations embargo (1990–2003), Sadam Hussain's government used its control over food markets to reward cronies and punish opponents, thereby actually improving its hold over the country. Similarly, Jones (2015) finds that the imposition of sanctions enabled political actors not only in Iraq, but also in Libya, Sudan, and Myanmar, to leverage their control over black-market activities, increasing their benefits from illegal gains as a result of the sanctions, even compared to the revenues from the legal export of the same goods previously. While these issues – as we discussed in detail in the theory section – are possible and as such deserve future research, as we also explained above, they will drive our estimates downward (that is to zero), and as such, provide – if anything – for a more stringent test of our theory.

The final scope condition relates to the impact of interstate and extrasystemic conflicts. Our analysis focused on violence occurring during civil war, the most prevalent conflict form since WWII (Sundberg & Melander, 2013). While we cannot conclude that the relationships we identify are valid for interstate conflict contexts, anecdotal evidence suggests similar substitution dynamics may occur in countries embroiled in interstate conflict. For example, the U.S. Secretary of State Anthony Blinken (2023) recently acknowledged that during the course of the Russia-Ukraine war "globally coordinated sanctions have forced the Russian Federation to search for alternate routes to finance and fuel its war machine," as oil export bans have forced Russia to sell crude oil at discounted prices (e.g., Mukherji, 2023) and the cost of importing banned products (e.g., Western technology) from third parties has sharply increased (Swanson & Stevis-Gridneff, 2023). One method of sanctions evasion, according to numerous reports, includes the violent seizure of significant quantities of grain from Ukraine (e.g., Lister & Fylyppov, 2022), which are marketed as Russian grain products (i.e., goods that are not subject to Western sanctions) and sold globally for profit. Walsh and Hopkins (2022), for instance, found that "Russia ha[d] stolen up to 500,000 tons of Ukrainian wheat, worth $100 million," much of it "taken from storage elevators in occupied parts of the Zaporizhzhia, Kherson, Donetsk and Luhansk regions," and a more recent report uncovered a massive smuggling operation, which seized "Ukrainian grain worth at least $530 million — cash that has helped feed President Vladimir Putin's war machine" (Biesecker, El Deeb, & Dupuy, 2022). This evidence suggest that further exploration might be able to robustly link our theory to sanction dynamics related to interstate and extrasystemic conflicts.

Finally, as one reviewer correctly noted, as sanctions status is not randomly assigned, any group characteristic correlated with sanction status could, potentially, explain the observation that this specific status is associated with actors committing more violence against civilians in cash-crops areas during more fertile periods. Following Occom's Razor logic, we believe that the most effective explanation as to why intensification of violence by sanctioned groups is sensitive to cash crop abundance is their viability as sources of revenue combined with the fact that they are produced by civilians who also depend on these crops. We also sought to empirically account for at least some of the more likely alternative explanations in our models (including conflict seasonality), while providing qualitative evidence to illustrate the viability of this argument in our case study. At the same time, we recognize that these observed trends could be explained by other factors related to sanctions, including these actors' territorial base, or general seasonality of fighting for the predominant actors in the sample. Future work could test the viability of these arguments against the one we advance here, or assess the possibility of several of these explanations working simultaneously.

7. Conclusion

For researchers, the findings suggest that studying the subnational

and subannual impacts of targeted sanctions is necessary for understanding the true magnitude of their second-order effects. Our focus on violence against civilians provides one, especially pernicious illustration, but there numerous other relevant subnational and subannual phenomena that can be affected by sanctions, including, among others, state capacity, public health, and economic and food security. For policymakers, the findings suggest that despite being utilized to minimize human suffering, targeted sanctions can have dire humanitarian effects if and when armed actors have an opportunity to violently seize lucrative agricultural goods.

With these potential second-order effects of sanctions in mind, our theory and findings provide useful guidance for how to inform the design and implementation of targeted sanctions regimes to both reduce the risk to civilians and increase the pressures on the targeted actors. Nations and international organizations who employ active sanctions regimes on specific actors can, for instance, send peacekeepers or military forces to provide security in cash-crop producing locations during agriculturally abundant times. They could also consider imposing additional sanctions on their targets' alternative sources of income, namely cash crops, especially those that do not provide necessary sustenance to locals (e.g., coffee, tea, qat). However, if the risk of food insecurity increases as a result of these measures, donors should supplement sanctions with food aid, which has been found to save lives and reduce suffering in conflict-torn countries (e.g., Mary & Mishra, 2020). Taking these steps can increase the success rate of targeted sanctions while reducing their adverse effects on civilians.

CRediT authorship contribution statement

Jerry Urtuzuastigui: Conceptualization, Data curation, Writing – original draft, Writing – review & editing. **Ore Koren:** Conceptualization, Data curation, Funding acquisition, Methodology, Project administration, Supervision, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data are openly available at: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/TI04S5.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.worlddev.2024.106606.

References

Addison, T., Le Billon, P., & Mansoob, S. (2002). Conflict in Africa: The cost of peaceful behaviour. *Journal of African Economies, 11*(3), 365–386.
Ahram, A. I. (2023). The paperwork of plunder: How rebels govern illicit resources. *International Politics*, 1–22.
Allendoerfer, M. G., Murdie, A., & Welch, R. M. (2020). The path of the boomerang: Human rights campaigns, third-party pressure, and human rights. *International Studies Quarterly, 64*(1), 111–119.
Angrist, J. D., & Pischke, J. S. (2009). *Mostly Harmless econometrics*. Princeton, NJ: Princeton University Press.
Asal, V., Phillips, B. J., Karl Rethemeyer, R., Simonelli, C., & Young, J. K. (2019). Carrots, sticks, and insurgent targeting of civilians. *Journal of Conflict Resolution, 63*(7), 1710–1735.
Azam, J.-P., & Hoeffler, A. (2002). Violence against civilians in civil Wars: Looting or terror? *Journal of Peace Research, 39*(4), 461–485.
Bahadur, J. 2022. "Terror and Taxes: Inside al-Shabaab's Revenue-Collection Regime." Global Initiative Against Transnational Terrorism. Accessed January 16, 2023. https://globalinitiative.net/wp-content/uploads/2022/12/AS-protection-economies.-WEB.pdf.
Bannon, I., & Collier, P. (2003). *Natural resources and violent conflict: Options and actions*. Washington, D.C.: World Bank.
Bapat, N. A., Heinrich, T., Kobayashi, Y., & Clifton, T. (2013). Determinants of sanctions effectiveness: Sensitivity analysis using new data. *International Interactions, 39*(1), 79–98.
Barasa, L. 2014. Kenyan troops restore order to Somalia. Sunday Nation (Kenya). January 26, 2014. https://advance-lexis-com.proxyiub.uits.iu.edu/api/document?collection=news&id=urn:contentItem:5BCB-HS91-JDJN-60T3-00000-00&context=1516831.
Biersteker, T. (2014). Targeted sanctions consortium database codebook Accessed February 1, 2023. https://www.graduateinstitute.ch/sites/internet/files/2019-10/TSC%20Database%20Codebook.pdf
Biersteker, T. J., Eckert, S. E., Tourinho, M., & Hudáková, Z. (2018). UN targeted sanctions datasets (1991–2013). *Journal of peace research, 55*(3), 404–412.
Biersteker, T. J, Sue, E. E., Marcos, T. 2016. Targeted sanctions. Cambridge University Press.
Biesecker, M., Sarah, E. D., Beatrice, D. 2022. "Russia smuggling Ukrainian grain to help pay for Putin's war." Associated Press News. October 3, 2022. Accessed July 31, 2023.https://apnews.com/article/russia-ukraine-putin-business-lebanon-syria-87c3b6fea3f4c326003123b21aa78099.
Blinken, A. 2023. Further curbing russia's efforts to evade sanctions and perpetuate its war against Ukraine. United States State Department Press Statement. April 12, 2023. Accessed July 31, 2023. https://www.nytimes.com/2022/06/05/world/africa/ukraine-grain-russia-sales.html.
Bloomberg. 2023. Russia Chips Away at Painful Western Oil Sanctions. Newsletter. July 24, 2023. Accessed July 24, 2023. https://www.bloomberg.com/news/newsletters/2023-07-24/russia-chips-away-at-painful-western-oil-sanctions.
Boniface, B. 2014. Lamu attacks expose financial vulnerability of perpetrators. Sabahi (All Africa Global Media) July 31, 2014. https://advance-lexis-com.proxyiub.uits.iu.edu/api/document?collection=news&id=urn:contentItem:5CT8-SJ91-DY15-S264-00000-00&context=1516831.
Buba, I. (2023). Civilian protection payment and the escalation of violence against civilians in Northwestern Nigeria. *Global Studies Quarterly, 3*(2), ksad033.
Busetto, L., & Ranghetti, L. (2016). MODIStsp: An R package for automatic preprocessing of MODIS land products time series. *Computers & Geosciences, 97*.
Butler, C. K., & Gates, S. (2012). African range wars: Climate, conflict, and property rights. *Journal of Peace Research, 49*(1), 23–34.
Carneiro, C. L., & Jr, L. A. (2016). Targeted versus conventional economic sanctions: What is at stake for human rights? *International Interactions, 42*(4), 565–589.
Lucena, C. C., Laerte, A. Jr. 2016. Targeted versus conventional economic sanctions: what is at stake for human rights? International Interactions 42(4):565–589.
Collier, P., & Hoeffler, A. (2004). Greed and grievance in civil war. *Oxford economic papers, 56*(4), 563–595.
Cox, D. G., & Cooper, A. (2006). Democratic sanctions: Connecting the democratic peace and economic sanctions. *Journal of Peace Research, 43*(6), 709–722.
Crost, B., & Felter, J. H. (2020). Export crops and civil conflict. *Journal of the European Economic Association, 18*(3), 1484–1520.
Deutsche, P. 2012. FEATURE: Joy in rural Somalia as al-Shabaab loses ground. Deutsche Presse-Agentur. (June 2, 2012). https://advance-lexis-com.proxyiub.uits.iu.edu/api/document?collection=news&id=urn:contentItem:55SS-F5T1-JCKJ-J31Y-00000-00&context=1516831.
Drezner, D. W. (2011). Sanctions sometimes smart: Targeted sanctions in theory and practice. *International Studies Review, 13*(1), 96–108.
Drezner, D. W. (2015). Targeted sanctions in a world of global finance. *International Interactions, 41*(4), 755–764.
Early, B. R. (2015). *Busted sanctions: Explaining why economic sanctions fail*. Stanford University Press.
Early, B. R., & Cilizoglu, M. (2020). Economic sanctions in flux: Enduring chal- lenges, new policies, and defining the future research agenda. *International Studies Perspectives, 21*(4), 438–477.
Early, B., & Peksen, D. (2019). Searching in the shadows: The impact of economic sanctions on informal economies. *Political Research Quarterly, 72*(4), 821–834.
Early, B. R., & Peksen, D. (2022). Does misery love company? Analyzing the global suffering inflicted by US economic sanctions. *Global Studies Quarterly, 2*(2).
Early, B. R., & Schulzke, M. (2019). Still unjust, just in different ways: How targeted sanctions fall short of just war theory's principles. *International Studies Review, 21*(1), 57–80.
Eck, K. (2014). Coercion in rebel recruitment. *Security Studies, 23*(2), 364–398.
Elliott, K. A. (2016). *The impacts of United Nations targeted sanctions* (pp. 172–189). Targeted sanctions: The impacts and effectiveness of United Nations action.
Elliott, K. A., & Hufbauer, G. C. (1999). Same song, same refrain? Economic sanctions in the 1990's. *American Economic Review, 89*(2), 403–408.

Eriksson, M., & Wallensteen, P. (2015). Targeting sanctions and ending armed conflicts: First steps towards a new research agenda. *International Affairs, 91*(6), 1387–1398.
Escribà-Folch, A. (2010). Economic sanctions and the duration of civil conflicts. *Journal of Peace Research, 47*(2), 129–141.
Estancona, C. L. (2021). Rebel primary commodity markets, price shocks, and supplier victimization. *International Studies Quarterly, 65*(4), 1111–1123.
Fjelde, H. (2015). Farming or fighting? Agricultural price shocks and civil war in Africa. *World Development, 67*, 525–534.
Grauvogel, J., Licht, A. A., & von Soest, C. (2017). Sanctions and signals: How international sanction threats trigger domestic protest in targeted regimes. *International Studies Quarterly, 61*(1), 86–97.
Haggard, S., & Noland, M. (2023). *Economic Sanctions During Humanitarian Emergencies: The Case of North Korea | East-West Center |*.
Hendrix, C. S., & Young, J. K. (2014). State capacity and terrorism: A two- dimensional approach. *Security Studies, 23*(2), 329–363.
Herrera, J. S., & Martinez-Alvarez, C. B. (2022). Diversifying violence: Mining, export-agriculture, and criminal governance in Mexico. *World Development, 151*, Article 105769.
Hovil, L., & Werker, E. (2005). Portrait of a failed rebellion: An account of rational, sub-optimal violence in western Uganda. *Rationality and Society, 17*(1), 5–34.
Hultman, L. (2007). Battle losses and rebel violence: Raising the costs for fighting terrorism and political violence. *Terrorism and Political Violence, 19*(2), 205–222.
Hultman, L., & Peksen, D. (2017). Successful or counterproductive coercion? The effect of international sanctions on conflict intensity. *Journal of Conflict Resolution, 61*(6), 1315–1339.
Humphreys, M., & Weinstein, J. M. (2006). Handling and manhandling civilians in civil war. *American Political Science Review, 100*(3), 429–447.
Jaafar, H. H., & Woertz, E. (2016). Agriculture as a funding source of ISIS: A GIS and remote sensing analysis. *Food Policy, 64*, 14–25.
Jibril, D. 2013. Somali farmers relish freedom from Al-Shabaab extortion. Sabahi (All Africa Global Media). (August 27, 2013). https://advance-lexis-com.proxyiub.uits.iu.edu/api/document?collection=news&id=urn:contentItem:5977-KX31-DY15-S0G3-00000-00&context=1516831.
Jones, L. (2015). *Societies under siege: Exploring how international economic sanctions (do not) work*. New York: Oxford University Press.
Kaempfer, W. H., & Lowenberg, A. D. (1988). The theory of international economic sanctions: A public choice approach. *The American Economic Review, 78*(4), 786–793.
Kelley, J. G., & Simmons, B. A. (2015). Politics by number: Indicators as social pressure in international relations. *American Journal of Political Science, 59*(1), 55–70.
Koren, O., & Bagozzi, B. E. (2017). Living off the land: The connection be- tween cropland, food security, and violence against civilians. *Journal of Peace Research, 54*(3), 351–364.
Koren, O., & Schon, J. (2023). Climate change, cash crops, and violence against civilians in the Sahel. *Regional Environmental Change, 23*(3), 112.
Kustra, T. (2022). Economic sanctions as deterrents and constraints. *Journal of Peace Research, 00223433221088323*.
Lektzian, D., & Regan, P. M. (2016). Economic sanctions, military interventions, and civil conflict outcomes. *Journal of Peace Research, 53*(4), 554–568.
Levy, I., & Yusuf, A. (2021). How do terrorist organizations make money? Terrorist funding and innovation in the case of Al-Shabaab. *Studies in Conflict & Terrorism, 44*(12), 1167–1189.
Lister, T., Sanyo, F. 2022. Russians steal vast amounts of Ukrainian grain and equipment, threatening this year's harvest CNN. May 5, 2022. Accessed July 31, 2023. https://www.cnn.com/2022/05/05/europe/russia-ukraine-grain-theft-cmd-intl/index.html.
Lu, Y., & Yamazaki, S. (2023). Fish to fight: Does catching more fish increase conflicts in Indonesia? *World Development, 170*, Article 106337.
Lujala, P. (2009). Deadly combat over natural resources: Gems, petroleum, drugs, and the severity of armed civil conflict. *Journal of Conflict Resolution, 53*(1), 50–71.
Mangan, F., Acko, I., & Taha, M. (2020). "The 'green diamond:' coffee and conflict in the Central African Republic. *United States Institute of Peace*. https://www.usip.org/publications/2020/02/green-diamond-coffee-and-conflict-central-african-republic.
Markowitz, L. P. (2017). The resource curse reconsidered: Cash crops and local violence in Kyrgyzstan. *Terrorism and Political Violence, 29*(2), 342–358.
Martin-Shields, C. P., & Stojetz, W. (2019). Food security and conflict: Empirical challenges and future opportunities for research and policy making on food security and conflict. *World Development, 119*, 150–164.
Mary, S., & Mishra, A. K. (2020). Humanitarian food aid and civil conflict. *World Development, 126*, Article 104713.
McKenzie, D., Teo, K. CNN. 2012. Is narcotic khat funding terrorism?. CNN Wire. March 7, 2012. https://advance-lexis-com.proxyiub.uits.iu.edu/api/document?collection=news&id=urn:contentItem:554F-3XW1-DY7V-G548-00000-00&context=1516831.
McLean, E. V., & Whang, T. (2021). Economic sanctions and government spend- ing adjustments: The case of disaster preparedness. *British Journal of Political Science, 51*(1), 394–411.
Millán-Quijano, J., & Pulgarín, S. (2023). Oiling up the field. Forced internal displacement and the expansion of palm oil in Colombia. *World Development, 162*, Article 106130.
Mukherji, B. (2023). *India, China 'enjoy the discount' on Russian oil as EU tightens price cap*. South China Morning Post. https://www.scmp.com/week-asia/economics/article/3210640/india-china-enjoy-discount-russian-oil-eu-tightens-price-cap.
Neuenkirch, M., & Neumeier, F. (2016). The impact of US sanctions on poverty. *Journal of Development Economics, 121*, 110–119.
Ornelas, R. G. (2018). Organized crime in Michoacán: Rent-seeking activities in the avocado export market. *Politics & Policy, 46*(5), 759–789.
Pape, R. A. (1997). Why economic sanctions do not work. *International Security, 22*(2), 90–136.
Peksen, D. (2009). Better or worse? The effect of economic sanctions on human rights. *Journal of Peace Research, 46*(1), 59–77.
Peksen, D. (2019a). Political effectiveness, negative externalities, and the ethics of economic sanctions. *Ethics & International Affairs, 33*(3), 279–289.
Peksen, D. (2019b). When do imposed economic sanctions work? A critical review of the sanctions effectiveness literature. *Defence and Peace Economics, 30*(6), 635–647.
Peksen, D., & Cooper, A. (2010). Coercive or corrosive: The negative impact of economic sanctions on democracy. *International Interactions, 36*(3), 240–264.
Radtke, M., & Jo, H. (2018). Fighting the hydra: United Nations sanctions and rebel groups. *Journal of Peace Research, 55*(6), 759–773.
Salehyan, I., Siroky, D., & Wood, R. M. (2014). External rebel sponsorship and civilian abuse: A principal-agent analysis of Wartime atrocities. *International Organization, 68*(3), 633–661.
Schon, J., & Koren, O. (2022). Introducing AfroGrid, a unified framework for environmental conflict research in Africa. *Scientific data, 9*(1), 1–11.
Seawright, J., & Gerring, J. (2008). Case selection techniques in case study research: A menu of qualitative and quantitative options. *Political Research Quarterly, 61*(2), 294–308.
Stoelinga, N. (2023). Cultivation and competition in Colombia: Disentangling the effects of coca price changes on violence. *Journal of International Development*.
Stoop, N., Verpoorten, M., & Van Der Windt, P. (2019). Artisanal or industrial conflict minerals? Evidence from eastern Congo. *World Development, 122*, 660–674.
Sundberg, R., & Melander, E. (2013). Introducing the UCDP georeferenced event dataset. *Journal of Peace Research, 50*(4), 523–532.
Swanson, A., & Stevis-Gridneff, M. (2023). Russia Is Importing Western Weapons Technology, Bypassing Sanctions. *The New York Times*. https://www.nytimes.com/2023/04/18/business/economy/us-russia-chips-sanctions.html.
Sweet, R. (2021). Concealing conflict markets: How rebels and firms use state institutions to launder wartime trade. *International Organization, 75*(4), 1109–1132.
Tokdemir, E., & Mark, B. S. (2018). When killers become victims: Diversionary war, human rights, and strategic target selection. *International Interactions, 44*(2), 337–360.
Tollefsen, A. F., Strandand, H., & Buhaug, H. (2012). PRIO-GRID: A unified spatial data structure. *Journal of Peace Research, 49*(2), 363–374.
U.N. Security Council (UNSC). 2016. "Report of the Monitoring Group on Somalia and Eritrea pursuant to Security Council resolution 2244 (2015): Somalia (S/2016/919)".
U.S. Department of State (USDS). 2013. Somalia; Human Rights Reports - Somalia. April 19, 2013. https://advance-lexis-com.proxyiub.uits.iu.edu/api/document?collection=news&id=urn:contentItem:588P-83D1-JC86-C04K-00000-00&context=1516831.
U.S. Department of Treasury (USDT). 2023. "Treasury Designates Terror Operatives and Illicit Charcoal Smugglers in Somalia." Press Release. May 23, 2023. Accessed July 21, 2023. https://home.treasury.gov/news/press-releases/jy1499.
Ubilava, D., Hastings, J. V., & Atalay, K. (2023). Agricultural windfalls and the seasonality of political violence in Africa. *American Journal of Agricultural Economics, 105*(5), 1309–1332.
Urtuzuastigui, J. (2023). American diasporas, homeland human rights conduct, and the onset of human rights-based economic sanctions. *International Interactions*, 1–28.
U.S. Department of State (USDS). 2012. Somalia; Human Rights Reports - Somalia. https://2009-2017.state.gov/documents/organization/204377.pdf.
Uzonyi, G., & Demir, B. (2020). Excluded ethnic groups, conflict contagion, and the onset of genocide and politicide during civil war. *International Studies Quarterly, 64*(4), 857–866.
Van, W. S. 2016. Short term effects of drought on communal conflict in Nigeria (No. WP16/18). UCD Centre for Economic Research Working Paper Series. https://www.econstor.eu/bitstream/10419/175477/1/WP16_18.pdf.
Walsh, D., Valerie, H. 2022. "Russia smuggling Ukrainian grain to help pay for Putin's war." New York Times. June 5, 2022. Accessed July 31, 2023. https://www.nytimes.com/2022/06/05/world/africa/ukraine-grain-russia-sales.html.
Weinstein, J. M. (2005). Resources and the information problem in rebel recruitment. *Journal of Conflict Resolution, 49*(4), 598–624.
Whitaker, B. E., Walsh, J. I., & Conrad, J. (2019). Natural resource exploitation and sexual violence by rebel groups. *The Journal of Politics, 81*(2), 702–706.
Woertz, E. (2019). Iraq under UN embargo, 1990–2003: Food security, agriculture, and regime survival. *The Middle East Journal, 73*(1), 92–111.
Wood, R. M. (2008). 'A hand upon the throat of the nation': Economic sanctions and state repression, 1976–2001. *International Studies Quarterly, 52*(3), 489–513.
Wood, R. M. (2010). Rebel capability and strategic violence against civilians. *Journal of Peace Research, 47*(5), 601–614.
Wood, R. M. (2014). From loss to looting? Battlefield costs and rebel incentives for violence. *International Organization, 68*(4), 979–999.
Yu, Q., You, L., Wood-Sichra, U., Yating, Ru., Joglekar, A. KB., Fritz, S., … Yang, P. (2020). A cultivated planet in 2010–Part 2: The global gridded agricultural-production maps. *Earth System Science Data, 12*(4), 3545–3572.